

Time complexity

Exercise sheet 7

Exercise 7.1. Consider the following algorithm for primality testing. What is its worst-case running time in terms of the input length when N is written in unary, binary, or decimal?

```
def is_prime(N):
    for d in range(2, N):
        if N % d == 0:
            return False
    return (N >= 2) and True
```

Exercise 7.2. The input consists of binary integers $a, b \geq 0$ and $c > 0$. Design an algorithm that computes $a^b \bmod c$ using $O(\log(b+1))$ modular multiplications.

Exercise 7.3. A stack contains pancakes of n distinct sizes. One move reverses the top j pancakes, for a chosen j . Give an algorithm that sorts the stack in the increasing order using at most $2n - 2$ moves.

Exercise 7.4. The input consists of binary integers $N \geq 0$ and $m \geq 1$. Give an algorithm that computes $\lfloor N^{1/m} \rfloor$ in time polynomial in the lengths of the two inputs.

Exercise 7.5. Prove that the following problem 2COLOUR is in P:

Input: graph G ;

Question: does G admit a proper colouring in two colours?

Exercise 7.6. Prove that the following problem UNARY_SUBSET_SUM is in P:

Input: Target $t \in \mathbb{N}$ and numbers $a_1, \dots, a_m \in \mathbb{N}$, given in *unary*.

Question: Is there a set of indices $I \subseteq \{1, \dots, m\}$ such that $\sum_{i \in I} a_i = t$?

Exercise 7.7. For a language L , let $L^* = \{w_1 \cdots w_r \mid r \geq 0, w_i \in L\}$. Prove that $L \in \text{P}$ implies $L^* \in \text{P}$. Note that $\varepsilon \in L^*$ even if $\varepsilon \notin L$.

Exercise 7.8. * A *literal* is a Boolean variable x or its negation $\neg x$. A *2CNF formula* is a conjunction of clauses, each containing at most two literals. For example,

$$(x \vee \neg y) \wedge (y \vee z) \wedge (\neg z)$$

is a 2CNF formula. Let 2SAT be the problem of deciding whether a given 2CNF formula has a satisfying assignment. Prove that $2\text{SAT} \in \text{P}$.